

CASC 2023	Schedule		University of Havana	
	Monday 28	Tuesday 29	Wednesday 30	Thursday 31
9:00-9:30	Registration	<i>Session Chair: Chenqi Mou</i> Ayoola Jinadu, Michael Monagan Solving Parametric Linear Systems using Sparse Rational Function Interpolation	<i>Session Chair: Jing Yang</i> Hollie Baker Cylindrical Algebraic Decompositions with Frontier Condition	<i>Session Chair: Chenqi Mou</i> Valentin Ittegov, Tatiana Titorenko On the Qualitative Analysis of the Equations of Motion of a Nonholonomic Mechanical System
9:30-10:00		Alexander Brandt, Juan Pablo González Trochez, Marc Moreno Maza, Haoze Yuan A Modular Algorithm for Computing the Intersection of a One-Dimensional Quasi- Component and a Hypersurface	Rashid Barket, Matthew England, Juergen Gerhard Generating Elementary Integrable Expressions	Andre Galligo, Didier Clamond Certified Study of Internal Solitary Waves
10:00-10:30		Opening	Timur Sadykov Grothendieck Residues and Frobenius-Puiseux Polynomials as Generating Solutions to Holonomic Systems of Equations	Ling Tan, Bo Li, BingWei Zhang, Jin-San Cheng An optimal algorithm for the intersection problem of plane curves defined by rational parametrization
10:30-11:00	Coffee break			
11:00-11:30	Coffee break	<i>Session Chair: Chenqi Mou</i> Stephen Watt Efficient Quotients of Non- Commutative Polynomials	<i>Session Chair: Jin-san Cheng</i> Soo Go, Victor Y. Pan, Pedro Soto Root-Squaring for Root-Finding	<i>Session Chair: Timur Sadykov</i> Mizuki Yoshizawa, Akira Terui, Masahiko Mikawa Inverse Kinematics and Path Planning of Manipulator Using Real Quantifier Elimination Based on Comprehensive Gröbner Systems
11:30-12:00	<i>Session Chair: Akira Terui</i> Chenqi Mou, Qiye Song On the Reduced Gröbner Bases of Block Determinantal Ideals	George Labahn Normal Form of Integer Matrices	José Luis Crespo, Jaime Gutierrez, Angel Valle Random Number Generators Quality Assessment with Neural Networks	Roberto Mulet On the Performance of Local Search Algorithms for K-SAT Problems in Random Graphs
12:00-12:30	Siyan Deng, Gregory Reid Challenges in Symbolic-Numeric Computation for Differential Systems Arising from Symmetry		Elizabeth Kalinina, Alexei Uteshev, Marina Goncharova, Elena Lezhnina On the Distance to the Nearest Defective Matrix	
12:30-14:00	Lunch break			
14:00-14:30	<i>Session Chair: Jaime Gutierrez</i> A.A. Gusev, O. Chuluunbaatar, V.L. Derbov, R.G. Nazmtdinov, S.I. Vinitzky, P.W. Wen, C.J. Lin, H.M. Jia, L.L. Hai Symbolic-Numerical Algorithm for Solving the Problem of Heavy Ion Collisions in an Optical Model with a Complex Potential	<i>Session Chair: Katsusuke Nabeshima</i> Jacob Imre, David J. Jeffrey Non-principal Branches of Lambert W	Excursion	<i>Session Chair: Jonathan Klein</i> Alexander Bathkin, Zafar Khaydarov Strong Resonant Condition Computation and Formal Stability
14:30-15:00	Katsusuke Nabeshima, Shinichi Tajima Effective Algorithm for Computing Noetherian Operators of Positive Dimensional Ideals	Peter Olver On the Structure and Generators of Differential Invariant Algebras		Vladimir V. Kornyak Complementarity Principle and Description of Evolution in Constructive Quantum Mechanics
15:00-15:30	Kai Hormann, Chee Yap, Ya Shi Andrew Zhang Range Functions of Any Convergence Order and their Amortized Complexity Analysis	Mahsa Ansari, Michael Monagan Computing GCDs of Multivariate Polynomials over Algebraic Number Fields Presented with Multiple Extensions		Diogo Gomes, Friedemann Krannich, Bashayer Majrashi, Ricardo Ribeiro Algorithmic Detection of Conserved Quantities for Finite-Difference Schemes
15:30-16:00	Coffee break	Victor Edneral Integrable Cases of the Bautin System with Resonance in the Linear Part		Coffee break
16:00-16:30	<i>Session Chair: Bo Huang</i> Klara Bernauer, Clemens Hofstadler, Georg Regensburger How to Automate Proofs of Operator Statements: Moore- Penrose Inverse – a Case Study	Coffee break		<i>Session Chair: Jonathan Klein</i> Amir Hashemi, Daniel Lichtblau On the Complexity of Linear Algebra Operations over Algebraic Extension Fields
16:30-17:00	Bernhard Andraschko, Julian Danner, Martin Kreuzer SAT Solving Using XOR-OR-AND Normal Forms	<i>Session Chair: Elizabeth Kalinina</i> Evgenii Vorozhtsov A Symbolic-numeric Method for Solving the Poisson equation in Polar Coordinates		Business Meeting
17:00-17:30	Weidong Wang, Jing Yang Two Variants of Bezout Subresultants for Several Univariate Polynomials	Martin Kreuzer, Florian Walsh Computing the Binomial Part of a Polynomial Ideal		
17:30-18:00	Conference Photo	Akira Katayama Computing a Basis of Isogeny Groups Between Two Isogenous Ordinary Elliptic Curves over a Finite Field	Closing	
18:00-18:30	Welcome cocktail	K.H. Lee Fractals Generated by Modifying Aperiodic Substitution Tilings	Conference Dinner	
20:00-22:00				